

SABBIR AHMMED SHEHAB

ASSISTANT ENGINEER

Product Development | Industrial Automation | Engineering Documentation

+880 1635-166768 • sabbirahmmedshehab@gmail.com • Dhaka, Bangladesh



CORE EXPERTISE

- Engineering Product Development
- PLC & HMI Systems
- Embedded Systems & IoT
- Electrical & Electronics Trainers
- PCB Design & Prototyping
- Technical Documentation
- Manufacturing, Validation & QA
- Training, Handover & Customer Support

TECHNICAL TOOLKIT

PLC & Automation

Siemens LOGO! • Delta PLC • HMI • WPLSoft • Hardware configuration • Communication setup • Troubleshooting

Embedded & Electronics

Arduino • STM32 • ESP32/ESP8266 • PIC • Sensors • Power supplies • PCB assembly

Design & Simulation

Proteus • LTspice • EasyEDA • Adobe Illustrator • Photoshop

Documentation

User manuals • Experiment guides • Wiring diagrams • Installation & maintenance instructions • QR-enabled documentation

LANGUAGES

Bengali — Native

English — Strong writing; basic spoken communication

Hindi — Conversational

EDUCATION

B.Sc. in Electrical & Electronic Engineering (Ongoing)

Manarat International University

Diploma in Electronics Engineering

Rangpur Polytechnic Institute
CGPA: 3.35/4.00 | 2021

SSC, Science

Al-Hera Institute
GPA: 4.59/5.00 | 2017

PROFESSIONAL PROFILE

Assistant Engineer with 4+ years of hands-on experience developing and improving practical engineering trainer systems. Works across product requirements, component selection, PLC/HMI and embedded integration, panel and PCB assembly, functional testing, troubleshooting, production support and instructor handover. Strongest in turning technical concepts into buildable, testable products and supporting them through validation, delivery and customer use.

KEY ACHIEVEMENTS

- Developed or substantially upgraded 25+ trainer-system categories covering electrical, electronics, PLC/HMI, embedded systems, IoT, renewable energy, networking, robotics and computer hardware.
- Standardized QR-linked documentation across trainer boards, giving instructors and users faster access to manuals, wiring diagrams, experiments and troubleshooting guidance.
- Converted product requirements into repeatable production packages containing BOM inputs, panel layouts, wiring plans, experiments, test procedures and controlled handover documents.
- Delivered trainer operation, PLC, hardware and troubleshooting sessions across 9 institutions, including government polytechnics and PFI/Navy Base Kaptai.
- Supported government tender work through technical-specification review, supplier communication and sourcing of welding, SCARA and Cartesian robot solutions.

PROFESSIONAL EXPERIENCE

Assistant Engineer | Fabotronix Limited

February 2022 – Present | Dhaka, Bangladesh

- Contribute across product development, manufacturing, validation, training and customer-facing technical support as Assistant Engineer.
- Translate product and curriculum requirements into circuit concepts, component selection, BOM inputs, panel layouts, practical experiments and test procedures.
- Configure and troubleshoot Siemens LOGO! and Delta PLC/HMI systems for motor control, sensors, communication and industrial automation exercises.
- Coordinate with design, production, procurement, suppliers and instructors from prototype and assembly through design validation, inspection, delivery and technical handover.
- Perform panel wiring, PCB assembly, component installation, functional testing, root-cause troubleshooting, corrective action and final quality checks; maintain controlled manuals and QR-linked technical files.

Service Engineer | Gazi International

2023 | Six-month assignment

- Diagnosed, serviced and function-tested peripheral, centrifugal, submersible and drainage pumps, electric motors and selected home appliances.
- Handled disassembly, component inspection, fault isolation, repair, reassembly and customer-facing technical support; motor rewinding was outside the role.

KEY PRODUCT PORTFOLIO

Selected products developed, upgraded or production-supported:

Electrical & Measurement	Electronics & Embedded	Automation & Energy	ICT & Service Systems
• Electrical Measurement Trainer • DC/AC Circuit Trainer • Distribution & Protection Trainer	• Digital Logic Trainer • Microcontroller Trainer • Sensor & Industrial Electronics Trainer	• PLC & HMI Training System • Embedded IoT Trainer • Robotics / Conveyor Trainer • Solar & Hybrid Energy Trainer	• LAN & EPABX Trainer • PC / Laptop Hardware Trainer • LED TV Service Trainer

SELECTED ENGINEERING PROJECTS

PLC & HMI	Defined I/O architecture, control wiring, PLC exercises, communication setup, validation checks and instructor handover for Siemens LOGO! and Delta platforms.
Embedded & IoT	Integrated Arduino, STM32, ESP32/ESP8266, sensors and communication modules into modular trainer systems with practical experiments and troubleshooting points.
QR Documentation	Created a controlled QR-based structure for manuals, wiring diagrams, experiment guides and troubleshooting references, improving access during delivery and post-installation support.
Energy & Robotics	Supported solar, hybrid-energy, robotics and conveyor trainers through component selection, protection, wiring, assembly, functional testing and technical guidance.

MANUFACTURING & PRODUCTION CAPABILITY

Hands-on production Trainer and panel assembly • Panel wiring • PCB assembly • Component installation • BOM support • Design-for-assembly considerations	Testing & quality Functional testing • Design validation • Quality inspection • Test procedures • Root-cause analysis • Corrective troubleshooting • Final handover
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TECHNICAL TRAINING & KNOWLEDGE TRANSFER

9 Institutions Chapai Nawabganj • Dinajpur • Rangpur • Shariatpur • Sweden Polytechnic • PFI/Navy Base Kaptai • Sirajganj • Jashore • Satkhira	Training coverage Trainer operation • Experiments • Siemens LOGO! Soft Comfort • Delta WPLSoft • PLC hardware and communication • Troubleshooting • Pneumatic conveyor • LAN and logic-gate applications
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TRAINING, CERTIFICATION & EARLIER EXPERIENCE

Professional Development • Industrial Attachment: PCB Design, PLC & Microcontroller — Fabotronics (6 months) • Graphics Design & Microsoft Office — Zila Parishad Computer Training Center (6 months) • Technical service exposure — Gazi International (2023)	Earlier Engagements • Assistant Teacher — Protiva Bikash Kindergarten School (2018–2019) • Part-time Promotional Representative — OPPO (2017–2018) Additional strengths: cross-functional coordination, project planning, vendor communication, analytical problem solving, customer support and technical training.
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